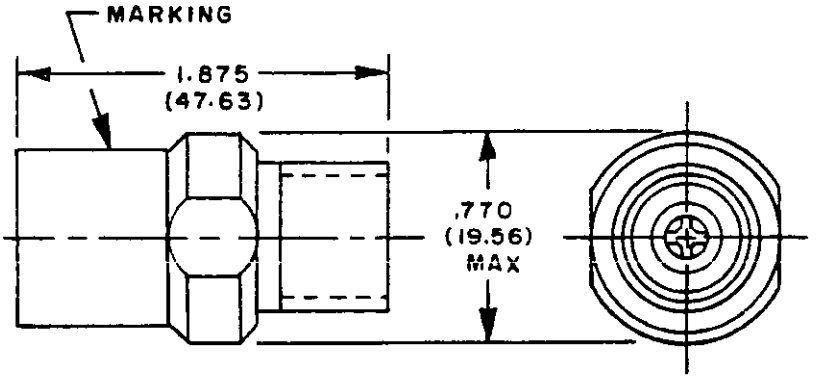
MILITARY SPECIFICATION SHEET

CONNECTORS, TRIAXIAL, RADIOFREQUENCY (SERIES TRT (CABLED) RECEPTACLE, SOCKET CONTACT, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

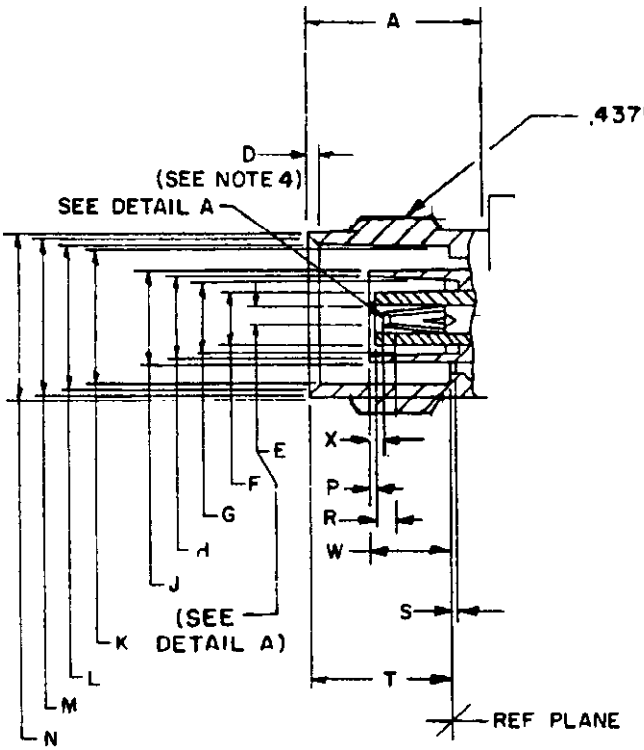
The complete requirements for acquiring the connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-49142.

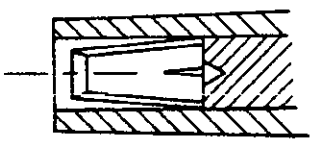


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Wrench flats are to accommodate standard wrench openings per FED-STD-H28.
5. 1.875 defines the maximum length of the connector when assembled to the appropriate table.
6. All undimensioned pictorial configurations are for reference purposes only.

FIGURE 1. General configuration.

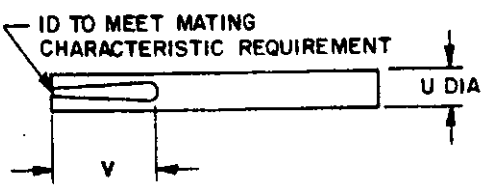
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ALTERNATE CLOSED ENTRY INSULATOR

DETAIL A



CONTACT

Ltr	INCHES		MILLIMETERS	
	Min	Max	Min	Max
A	.414	---	10.52	---
D	.015	.030	.38	.76
E	.039	.043	.99	1.09
F	.117	.122	2.97	3.10
G	.169	.171	4.29	4.34
H	.178	.182	4.52	4.62
J	.195	.199	4.95	5.05
K	.319	.321	8.10	8.15
L	.327	.333	8.31	8.46
M	.346	.356	8.79	9.04
N	.378	.382	9.60	9.70
P	.000	.015	.00	.38
R	.038	.050	.97	1.27
S	.001	---	.03	---
T	.327	.335	8.31	8.51
U	.062	.064	1.57	1.63
V	.213	---	5.41	---
W	.191	.213	4.85	5.41
X	.010	.038	.25	.97

NOTES:
1. Dimesions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are tabulated.
4. Chamfer or radius.

FIGURE 2. Mating dimensions for socket contact termination.

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ENGINEERING DATA:

Nominal impedance: Non constant.

Frequency range: 0 to 500 MHz, minimum.

Voltage rating: 400 V rms, maximum working voltage at sea level. 100 V rms, maximum working voltage at 70,000 feet.

Temperature range: -65° to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Force to engage and disengage:
Longitudinal force: Not applicable.
Torque: 2.5 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque - 4-6 inch-pounds.

Mating characteristics: See figure 2 for dimensions.
Center contact (socket):
Oversize test pin - 0.040 inch diameter, minimum (nonclosed entry contacts only).
Insertion depth: .125 inch, minimum.
Number of insertions - 1
Insertion force test:
Steel test pin finish - 16 microinches.
Insertion force - 2 pounds maximum.
Steel test pin diameter - .039 inch minimum.
Withdrawal force test:
Steel test pin diameter - .037 inch maximum.
Withdrawal force - 2 ounces, minimum.
Test pin finish - 16 microinches.

Permeability: Applicable.

Hermetic seal: Not applicable.

Leakage: Not applicable.

Insulation resistance: 5000 megohms.

Center conductor retention: 6 pounds minimum, axial force.

Dielectric withstanding voltage: At sea level, 1200 V rms, between center conductor and intermediate conductor. 500 V rms, between intermediate conductor and outer conductor.

Salt spray (corrosion): Applicable.

Vibration: Applicable.

Shock: Applicable.

Thermal shock: Applicable.

Moisture resistance: Applicable.

Conductor resistance: In milliohms, maximum.

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	Initial	After environment
Center conductor	2.0	2.5
Intermediate conductor	0.5	0.6
Outer conductor	0.2	0.3

TABLE I. Group Qualification.

Group	Submission and qualification of any of the following dash numbers	Qualifies the following dash numbers
I	-0001 -0002 -0007	-0001 -0002 -0007
II	-0003 -0004	-0003 -0004
III	-0005	-0005
IV	-0006	-0006

Corona level:
Altitude: 70,000 feet.
Voltage: 300 V rms, minimum

RF high potential withstanding voltage:
800 V rms, between center conductor and intermediate conductor.
200 V rms, between intermediate conductor and outer conductor at 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force (for cable .200 to .325 outside dimension): 40 pounds, minimum.

Coupling mechanism retention force: Not applicable.

Risetime degradation: To be added.

Connector durability: 500 cycles minimum, at 12 cycles per minute maximum.

Part number M49142/11-(dash number from table II).

TABLE II. Dash number, cross reference to cable.

Dash number	Cross reference number
0001	M17/134-00001
0002	M17/134-00002
0003	D3-7619-5/336
0004	D3-7619-5/338
0005	380-10045-1
0006 1/	M17/176-00002
0007	M17/116-RG307

1/ Not for Air Force use.

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Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 5935-3359-5)

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	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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